PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Machine Learning Techniques for Mortality Prediction in Emergency Departments: A Systematic Review
AUTHORS	Naemi, Amin; Schmidt, Thomas; Mansourvar, Marjan; Naghavi-Behzad, Mohammad; Ebrahimi, Ali; Wiil, Uffe

VERSION 1 – REVIEW

REVIEWER	Klang, Eyal Chaim Sheba Medical Center
REVIEW RETURNED	22-May-2021

GENERAL COMMENTS	The work “Machine Learning Techniques for Mortality Prediction in Emergency Departments: A Systematic Review” reviewed 10 papers that evaluated machine learning models for predicting mortality in the ED using vital signs data. The authors found methodological gaps and risk of bias in the reviewed studies. Thus, the authors formulated 8 recommendations for future similar research. The review is comprehensive, and the subject matter is important. Some remarks: 1) The authors state that: “Besides the vital signs, all studies used other clinical predictors such as demographic variables, blood tests, comorbidities, diagnoses, and triage information.“ a. I recommend that the authors expand on the other clinical features. b. Why did the authors choose “vital signs” as the research question. As stated, most studies used other features. Wouldn’t “triage data” be better suited as the research question? c. If most of the studies used other data such as labs, why only 10 studies were included? There are other studies that used ED vitals + other ED features to predict mortality. Some examples: i. Apr 2017, Duc T. Ha et al. Development and validation of prognostic model for predicting 30-day mortality risk in medical patients in emergency department (ED) ii. (BMJ open) Sep 2019, Feng Xie et al. Novel model for predicting inpatient mortality after emergency admission to hospital in Singapore: retrospective observational study iii. Epub 2020 Aug - Soffer et al. Predicting In-Hospital Mortality at Admission to the Medical Ward: A Big-Data Machine Learning Model 2) The authors gave 8 recommendations for future studies. a. Although rigorous research is important, some recommendations may limit the field. For instance, limiting research to prospective and externally validated data. b. The authors recommend that model building will use prospectively collected data. They also acknowledge that effective ML needs very

	large cohorts. These two requirements somewhat contradict each other. c. The authors recommend using two techniques: “iterative multivariate imputation” for missing data and SMOTE for unbalanced data. These are very specific recommendations that need to be supported by multiple rigorous studies. 3) “meaning the time that their models provide before mortality occurs.” – unclear sentence. 4) “only one study used information of each patient to predict the risk of mortality for the same patient” – can the authors expand on this statement? Supervised models are trained on one data-set and then predict labels of another data-set. Though done in batch, the prediction is made per row/patient, by only using the data of that row/patient. 5) The summary table of ML techniques is sketchy. Some examples: a. the authors state that LR is “not very accurate”. Yet, the authors also cited “A recent systematic review showed no performance benefits of ML methods over logistic regression.” b. The authors state some advantages of RF: “fast execution”, “useful for regression and classification”. Yet, GB implementation such as XGBoost is faster than RF, and may also be used for regression. c. What is the difference between NN (neural network) and DNN (deep neural network)? Does NN have only one hidden layer?

REVIEWER	Fernandes, Marta Massachusetts General Hospital
REVIEW RETURNED	31-May-2021

GENERAL COMMENTS	In this paper, the authors do a systematic review to assess the performance and clinical feasibility of ML algorithms in prediction of in-hospital mortality for medical patients using vital signs at emergency departments. The paper is easy to read and the methodology is well described to be reproducible. The major comment I indicate is to update the search for publication year and include more terms for ML. Comments Abstract – Conclusion Line 8: Replace “mature” by a more suited term such as “robust”. Introduction Line 16: Remove “algorithms” from “Unsupervised learning is subset of ML algorithms that” Line 25: Remove the comma at the end of this line. Section 3.5 Line 36: Replace “dead” by “non survived”, for consistency. Recommendation 6 Line 9: insert “a” in “is a challenge”. “CONSLUSION” is misspelled. Figures and Tables

	In Figure 1, remove “with reasons” where n=32. Table 2 – Did the authors considered adding more terms to group 1? There are other words related to ML which would allow extraction of more studies, such as: model, modeling, classification, prediction, classify, predict, predictive, predicting, classifying. I strongly advise to perform additional search with these and other terms the authors may find for Group 1 and adding the studies resulting from the search. I also advise to extend the search period for publication year from October 2020 to May 2021, to get the most updated results. Tables 4 and 5 can be part of the Appendix and 2 figures can be created from this tables instead, and presented in the main text of the manuscript.

VERSION 1 – AUTHOR RESPONSE

Reviewer 1: Dr. Eyal Klang		Response
1	The authors state that: “Besides the vital signs, all studies used other clinical predictors such as demographic variables, blood tests, comorbidities, diagnoses, and triage information.“ a. I recommend that the authors expand on the other clinical features. b. Why did the authors choose “vital signs” as the research question. As stated, most studies used other features. Wouldn’t “triage data” be better suited as the research question? c. If most of the studies used other data such as labs, why only 10 studies were included? There are other studies that used ED vitals + other ED features to predict mortality. Some examples: i.Apr 2017, Duc T. Ha et al. Development and validation of prognostic model for predicting 30-day mortality risk in medical patients in emergency department (ED) ii.(BMJ open) Sep 2019, Feng Xie et al. Novel model for predicting inpatient mortality after emergency admission to hospital in Singapore: retrospective observational study iii.Epub 2020 Aug - Soffer et al. Predicting In-Hospital Mortality at Admission to the Medical Ward: A BigData Machine Learning Model	a. Thanks for your suggestion. in section 3.2, we have added some of the other features used in the studies. b. Thanks for your comment. We investigated studies that have vital signs among their datasets’ features. We had some reasons for this: (1) vital signs are the cheapest, fastest, and easiest data can be collected at ED, (2) in almost all EDs, patients’ vital signs are among the first data that clinicians observed to have an initial assessment about the patients’ condition, (3) almost all triage systems utilize vital signs to stratify patients into different categories, (4) a recent study discussed: “vital signs play an important role in ED to identify patients at risk of deterioration. Even though it is accurately predicted by vital sign changes, clinical deterioration often goes unnoticed, or is not detected until it is too late to treat. This is mainly caused by inadequate recording of vital signs or as a result of an inappropriate response to abnormal values. Among nurses and doctors there is insufficient knowledge and appreciation of vital sign changes and their implications for patient care.” (doi.org/10.1371/journal.pone.0210875). This and similar studies motivated us to investigate the studies that had vital signs among their datasets’ features and utilized ML for mortality prediction. (5) in line with previous reasons, our colleagues at Odense University Hospital and we (Health Informatics Unit at University of Southern Denmark) are working on vital signs data

		and use them to predict adverse events using ML that can be clinical interpretable and meaningful, so this systematic provides an updated view about the current status of using ML and vital signs at EDs. For these reasons vital signs are among the most important and informative data that can be collected at EDs, so as the aim of presenting ML models is to be feasible in clinical practice, vital signs cannot be denied. We also reflect your comment in Limitations section of our study and recommend in future studies other features are also investigated. c. i., ii. Thanks for mentioning this type of studies. However, these studies did not mention machine learning and its applications at all. They utilized a R package to extract the most important variables related to mortality, and to do this they used conventional statistical analysis. However, after a discussion with our ML expert, we decided to exclude this type of studies because these models' aim is different from the aim of using ML to be used in clinical practice especially at ED for mortality prediction. iii. Thanks for suggesting this paper. After re-running our queries for years 2020 and 2021, we extracted 567 articles, and after screening, this paper was among the eligible papers.
2	The authors gave 8 recommendations for future studies. a. Although rigorous research is important, some recommendations may limit the field. For instance, limiting research to prospective and externally validated data. b. The authors recommend that model building will use prospectively collected data. They also acknowledge that effective ML needs very large cohorts. These two requirements somewhat contradict each other. c. The authors recommend using two techniques: "iterative multivariate imputation" for missing data and SMOTE for unbalanced data. These are very specific recommendations that need to be supported by multiple rigorous studies	a. Thanks for the nice reminder. We understand that is a rigorous way for validation, however external validation is still a common approach for showing the robustness of models. We added a reference for this to the paper. b. Thanks for your kind reminder. Maybe "instead of" in "recommendation 1" is not a good word and we have changed it to "besides". We understand it seems difficult, but prospective setting and external validation are the common approaches to validate the model. c. Thanks for the comment. there are some studies such as (doi.org/10.1093/ije/dyu080) (we have added this reference to the text-recommendation 3, ref 62), discussed that multivariate imputation has substantial advantages, as it allows missing data to be handled in a way that is unbiased and statistically valid. Also, we mentioned SMOTE as an example for balancing data, however it is one of the popular and very effective oversampling technique used in various fields of research. SMOTE providers checked its ability on nine datasets and their article has 15,217 citations. Moreover, various extension of SMOTE for various

		aims and circumstances were implemented.
3	“meaning the time that their models provide before mortality occurs.” – unclear sentence.	Thanks for mentioning this important point. We changed it to: “meaning the time that their models provide clinicians to intervene before mortality occurs.”
4	“only one study used information of each patient to predict the risk of mortality for the same patient” – can the authors expand on this statement? Supervised models are trained on one data-set and then predict labels of another data-set. Though done in batch, the prediction is made per row/patient, by only using the data of that row/patient.	Thanks for the nice comment. In most studies only one model for the whole dataset was built, and single or few measurements of each patient was collected, e.g. only one measurement of heart rate for each patient, so patients were not monitored over time. However, one study (A9) considered vital signs of patients over time and paid attention to the trajectories and changes in related values. We revised section 3.8.
5	The summary table of ML techniques is sketchy. Some examples: a. the authors state that LR is “not very accurate”. Yet, the authors also cited “A recent systematic review showed no performance benefits of ML methods over logistic regression.” b. The authors state some advantages of RF: “fast execution”, “useful for regression and classification”. Yet, GB implementation such as XGBoost is faster than RF, and may also be used for regression. c. What is the difference between NN (neural network) and DNN (deep neural network)? Does NN have only one hidden layer?	We agree that this needed to be further improved. We just wanted to provide a brief introduction and overview about advantages and disadvantages of various ML techniques mentioned in the paper. So, we replaced “complete” term by “brief” in section 3.6. We also checked and revised Table. 6, especially those areas mentioned by Dr. Eyal Klang: a. In section 4.1, we explained that poor performance of ML in clinical applications as well as using some techniques such as LR and DT (easy to implement and no needed tuning) can be due to lack of enough attention to some steps such as data preparation and models hyperparameters tuning for novel and more effective models. In section 4.2 which is written from clinical point of view (the view of clinicians about the utilization of ML at ED), we mentioned that systematic review that claimed LR can work better than other ML models (doi.org/10.1016/j.jclinepi.2019.02.004). To avoid misunderstanding, we have added sentence after “A recent systematic review…” in section 4.2 to relate this section and mentioned explanation in section 4.1. added sentence: “However, as mentioned in the previous section, this conclusion could be due to lack of enough attention to the important stages and practical issues such as data preparation or models’ tuning, …”. b. Very true, GB and its extension are classified as fast learners and can be used for both classification and regression tasks. Related phrases are added to the Table. 6: “fast execution” and “useful for regression and classification” c. Yes, this needed to be clarified. Considering feedforward architecture, usually having two or more

		hidden layers counts as DNN. However, there are some training algorithms developed only for DNNs. Moreover, there are some architectures (besides feedforward) belonging to DNNs. We have revised related fields in the Table. 6.
Reviewer 2: Dr. Marta Fernandes		Response
1	General comment: In this paper, the authors do a systematic review to assess the performance and clinical feasibility of ML algorithms in prediction of in-hospital mortality for medical patients using vital signs at emergency departments. The paper is easy to read and the methodology is well described to be reproducible. The major comment I indicate is to update the search for publication year and include more terms for ML.	It is related to comment 8, we have answered it in detail there
2	Abstract – Conclusion Line 8: Replace “mature” by a more suited term such as “robust”.	Thanks for the suggestion. We changed it.
3	Introduction Line 16: Remove “algorithms” from “Unsupervised learning is subset of ML algorithms that” Line 25: Remove the comma at the end of this line	Thanks for the suggestion. We removed that comma. (comma after “outperformed”)
4	Section 3.5 Line 36: Replace “dead” by “non survived”, for consistency.	Thanks for the nice suggestion. We have updated the phrase.
5	Recommendation 6 Line 9: insert “a” in “is a challenge”.	It was revised.
6	“CONSLUSION” is misspelled.	Thanks for the nice reminder. We corrected this.
7	Figures and Tables In Figure 1, remove “with reasons” where n=32.	Thanks, we removed it.
8	Table 2 – a. Did the authors considered adding more terms to group 1? There are other words related to ML which would allow extraction of more studies, such as: model, modeling, classification, prediction, classify, predict, predictive, predicting, classifying. I strongly advise to perform additional search with these and other terms the authors may find for Group 1 and adding the studies resulting from the search. b. I also advise to extend the search period for publication year from October 2020 to May 2021, to get the most updated results	a. Thanks for the nice comment. Initially, we had some similar terms such as “predict*” or “forecast*” and so on. These terms are general terms that lead to huge number of irrelevant studies, however our ML expert and librarian recommended us to avoid these general terms because the aim of using ML in every study is to predict or forecast or detection or classification and so on, so it is not necessary to have them among our queries. Instead, considering the MeSH hierarchy of Medline and Embase, we extract all existed ML terms and used them in our queries both as MeSH and body text terms. To have a brief and clean table, we did not provide all of the terms in table 2, but we provided them in our PRISMA checklist during the submission process. We also upload it as “search_strategy.docx” file as supplementary materials. b. Thanks

		for the nice suggestion. We ran our queries again for years 2020 and 2021. We extracted 567 articles and after title, abstract, and full-text screening, the included papers set was updated. Thus, we have extracted related item (based on Table 4, Table 5, and Table 7) and updated the manuscript.
9	Tables 4 and 5 can be part of the Appendix and 2 figures can be created from this tables instead, and presented in the main text of the manuscript.	Thanks for your suggestion. We discussed finding a way to provide informative figures out of these tables, however since almost all information in the tables is categorical, it seems difficult to provide one figure for each table, and it would result in several additional figures. Also, as we checked the literature, most of systematic reviews provided their main results as tables.

VERSION 2 – REVIEW

REVIEWER	Klang, Eyal Chaim Sheba Medical Center
REVIEW RETURNED	03-Sep-2021

GENERAL COMMENTS	This is a nice paper that addresses an important topic. The authors addressed all my questions satisfactorily. Good luck.